

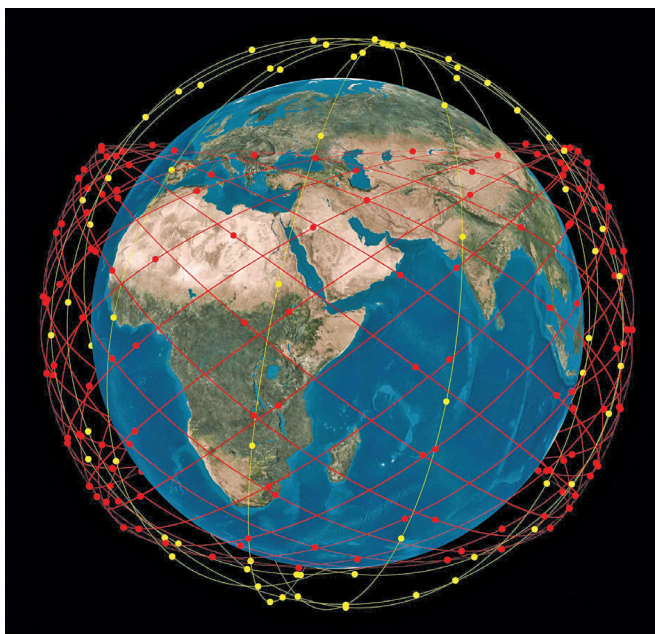
actual commercial constellation of LEO satellites was Iridium, launched in 1998. LEOs have now come under the limelight, thanks to SpaceX Starlink, which currently has more than 7500 LEO satellites in space, with plans to scale up to 42,000.

GEO. GEO satellites remain in fixed positions over the Earth's equator. Since they are the farthest away from the Earth, they have the widest coverage, and it is possible to cover almost the entire globe (except the extreme polar regions) with just three satellites. They also have a long operational life of more than 15 years. As GEO satellites are almost stationary, ground antennae do not require tracking capabilities and are very simple. However, being so far away, they are expensive to launch and consume a lot of power to communicate with ground stations. It is also challenging to deorbit these satellites, and almost impossible to repair!

The greatest disadvantage of this distance is the high signal latency, which makes GEO unsuitable for real-time applications. However, it remains the go-to choice for applications such as monitoring and broadcast services and connectivity for bank branches, ATMs, and other critical infrastructure, particularly in remote and underserved areas, due to its reliability.

LEO. As LEO satellites are closer to the Earth, it is less expensive to launch them. They also have lower power requirements and are easier to service or deorbit. This makes LEO constellations and communication systems scalable and cost-effective.

The proximity to Earth also



Telesat's LEO constellation of 120 small satellites (Source: Telesat)

gives LEO satellites their biggest advantage—a low latency (< 50ms), which provides connectivity akin to terrestrial broadband. This makes LEO satellite systems a perfect choice for broadband services, Internet of Things (IoT), mapping, environmental monitoring, and disaster management, which require real-time communications and high-resolution imaging.

However, being closer to the Earth also has its disadvantages. Each satellite covers only a limited area, which makes it necessary to have extremely large constellations for continuous global coverage. Also, due to the greater effect of atmospheric drag and gravitational variations, satellites need frequent station-keeping manoeuvres and orbital adjustments to keep the satellites within a desired orbit. The lifespan of a satellite is also significantly shorter, typically requiring replacement every 5 to 10 years.

Individual LEO satellites also experience problems with throughput because their small orbits provide a limited view-time of the

ground stations, which restricts the duration of data downlink and can cause potential data loss when transitioning from one satellite to another. Modern satellites attempt to overcome this limitation with inter-satellite links (ISLs) using laser beams, which effectively 'fill the gaps', as well as other methods such as link adaptation, optimised routing, and strategic ground station placement. Consequently, the aggregate throughput of a LEO constellation is terrific. LEO mega-constellations act like a cellular network in the sky, creating

immense capacity density in each area.

MEO. As with all 'medium' things, MEO falls between LEO and GEO in terms of cost, life, coverage, and latency. Although the transmitters need to be more powerful than LEO due to the longer distance, the ground systems require only moderate tracking capability. MEO satellite orbits are relatively stable and require less station-keeping manoeuvres. Although the latency of MEO satellites is higher than LEO, its coverage is also greater—a good balance, which makes it a popular choice for navigation systems. One caveat with MEO satellites is that they need radiation-hardened components as they pass through Van Allen radiation belts.

Each orbit has its pros and cons and is suited for specific applications and hence, it is impossible to say one is better than the other. LEO is ideal for low-latency communications and high-resolution imaging, while MEO is a good choice for navigation systems, and GEO for broadcasting and continuous